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How smartphone technology will evolve

While several technologies have evolved dramatically over time, few fields have witnessed the revolution seen in the smartphone technology landscape. One of the greatest 'equalizers' in history, they have ushered in a new era of human evolution. Today, they are the most affordable, convenient, and widely used internet-enabled devices in use across the world. If we were to look ahead to the future, we envision smartphones that are smarter and even more seamlessly integrated into our personal, professional and social lives. While the possibilities are endless, there are certain aspects that we can be fairly confident will develop along expected lines.

The smartphone has pretty much replaced traditional cameras for most people. It is estimated that the worldwide shipment of cameras dropped by almost 84% over the past decade, because the increase in quality of smartphone camera lenses and image sensors has enabled richer, sharper pictures and videos. In the future, you will be able to select from an even wider range of views, effects, and enhancements than today. These features will be based on technologies such as Augmented Reality, Virtual Reality, Artificial Intelligence, and more.

With selfies expected to remain popular, the front-facing camera will be accommodated underneath the display. Despite the challenges the industry has faced trying to accomplish this, it's only a matter of time before it becomes a reality.

The increasing consumption of video content on smartphones is driving the demand for larger displays – a trend that will likely continue in future. Entertainment aside, the use of smartphones for a wide range of gaming and work applications will drive further innovations around foldable, rollable, and interactive smartphone screens. We might even see holographic displays, which will be used with mixed-reality headsets to create breathtaking visual experiences for users.

With projected augmented reality likely to take center stage, interacting with multiple screens surrounding us regardless of size will take precedence. Creating a seamless blend of the physical and virtual worlds, these devices will be integrated into our daily lives.

People tend to use their smartphones a lot when they are on the move, and we expect wireless charging technologies to enable users to charge their device over the air at blazing speeds when they are within a few feet of a power transmitter.

The adoption of digital assistants such as Alexa, Cortana, Siri, Google Now, and many more is likely to grow exponentially over the coming decades as next-generation wireless networks and IoT-enabled devices become mainstream.

It is also likely that the future will see greater interplay between smartphones and IoT devices which might even lead to smartphones ceasing to exist as we know them. The power of a smartphone lies in the connectivity it provides and the next-generation wearable technologies are expected to evolve in capabilities to surpass smartphone technology in aspects like functionality, mobility, and processing. Wearable technologies are assuming different forms (bands, watches, clothing, glasses, etc.). In the future, smart watches and TWS may take over as the primary device for daily essential functions such as making calls. And smart screens could be built into every room, car and office. Imagine your refrigerator could read out your to-do list in the morning, your car's heads up display will show you the breakfast places on your route, and your AR glasses will let you join in on video calls at work – all of this and more, without carrying a physical smartphone device with you.

Meanwhile, mobile networks will evolve from 5G to 6G to 7G and beyond. In another couple of decades, Li-Fi will presumably have become as commonplace as Wi-Fi is today. Smartphone technology will evolve to support these networks and the myriad applications that they will enable. Device security will need to be at its strongest to ensure data privacy wherein we would be experiencing a much more complex yet evolved cybersecurity landscape. Meanwhile, amidst the sharpening focus on sustainability, manufacturers will explore unconventional, eco-friendly materials for designing smartphones and other smart devices, so as to minimize carbon footprint while delivering on all the necessary technical and functional requirements.

While the predictions are manifold, smartphones of the future may not even be smartphones at all. These devices as we currently use them, may take on an entirely new avatar while unlocking new functions and capabilities than ever witnessed before.

There is neither a crystal ball nor a prophecy that could tell us what the evolution of smartphone technology will hold. Surprise, it seems, will be the only constant.